Case 25: Spatially varying moisture quadrants

1 Verification purpose

This case verifies spatial moisture lookup, moisture damping, and extinction behavior in otherwise uniform fuel. The verification question is whether the four regional spread rates meet their moisture-specific references, including negligible spread at the extinction condition.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

Fuel moisture reduces reaction intensity through the model's moisture-damping term; at or above the fuel's moisture of extinction, sustainable spread approaches zero. Only the one-hour moisture raster varies, so regional velocity differences should follow that damping response.

For a dead-fuel moisture ratio $r = M_f/M_x$, the Rothermel damping polynomial used in the configured range is

$$\eta_M(r) = 1 - 2.59r + 5.11r^2 - 3.52r^3, \qquad 0 \le r \le 1. \tag{1}$$

Reaction intensity contains this damping factor, $I_R = \Gamma'(w_{n,d}\eta_{M,d} + w_{n,l}\eta_{M,l})\eta_s h$, and the no-wind rate is $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$. At or above extinction, the dead-fuel contribution cannot sustain propagation.

3 Derivation of expected behavior

For Fuel Model 3 the configured moisture values 6, 12, 18, and 25 percent correspond to reference rates 1.511, 1.089, 0.801, and $0\ \mathrm{m\,min^{-1}}$. The rates must decrease monotonically as moisture rises.

The dead one-hour moisture ratio is varied while all other generated fuel and moisture properties remain fixed. Applying $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$ and $Q_{ig} = 250 + 1116M_f$ gives

M_{1h}	I_R (kW m^{-2})	R_0 (m min^{-1})
0.06	549.174	1.511830
0.12	479.466	1.089719
0.18	413.934	0.801066
0.25	$\simeq 3.9 \times 10^{-13}$	$\simeq 6.5 \times 10^{-16}$

so the report references are $1.511,\ 1.089,\ 0.801$, and $0\ \mathrm{m\,min^{-1}}$. Nonzero rows use $e_R = |R_{\max} - R_{\mathrm{ref}}|/R_{\mathrm{ref}}$; the extinguished row uses the absolute residual $|R_{\max}|$, avoiding division by a zero reference.

4 Verification metrics and acceptance criteria

The maximum finite spread rate in each quadrant measures the local head response without averaging across moisture boundaries. Nonzero targets use 10 percent relative tolerance. The extinction quadrant

uses an absolute $0.05\ \mathrm{m\,min^{-1}}$ limit because relative error is undefined at zero. All four components must pass.

5 Simulation configuration and predicted outcome

The grid uses uniform Fuel Model 3, flat terrain, zero wind, central ignition, and one-hour moisture quadrants of 6, 12, 18, and 25 percent. Predicted propagation slows monotonically and stops in the 25-percent quadrant.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

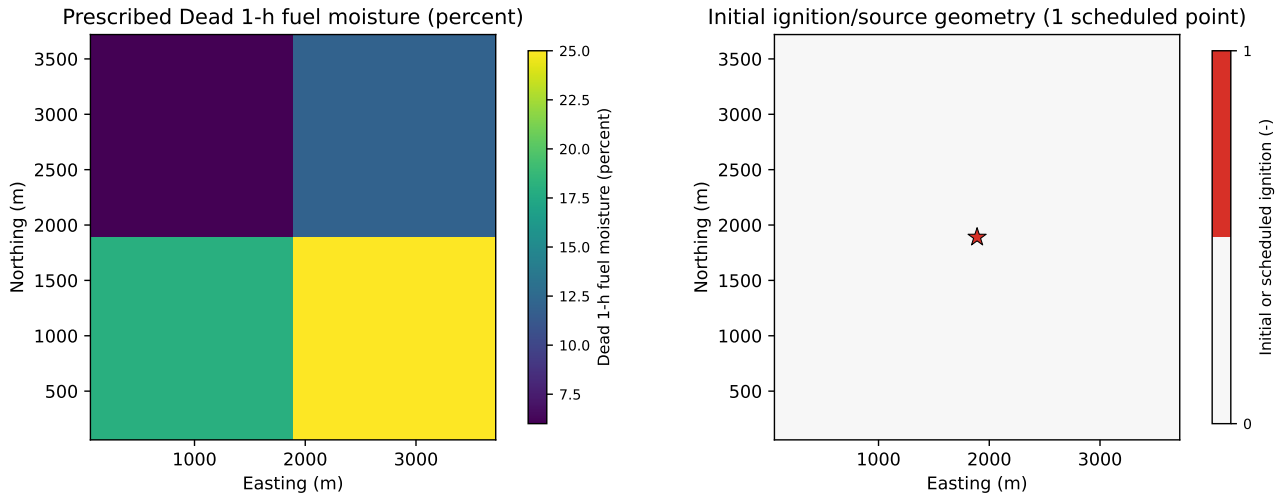


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

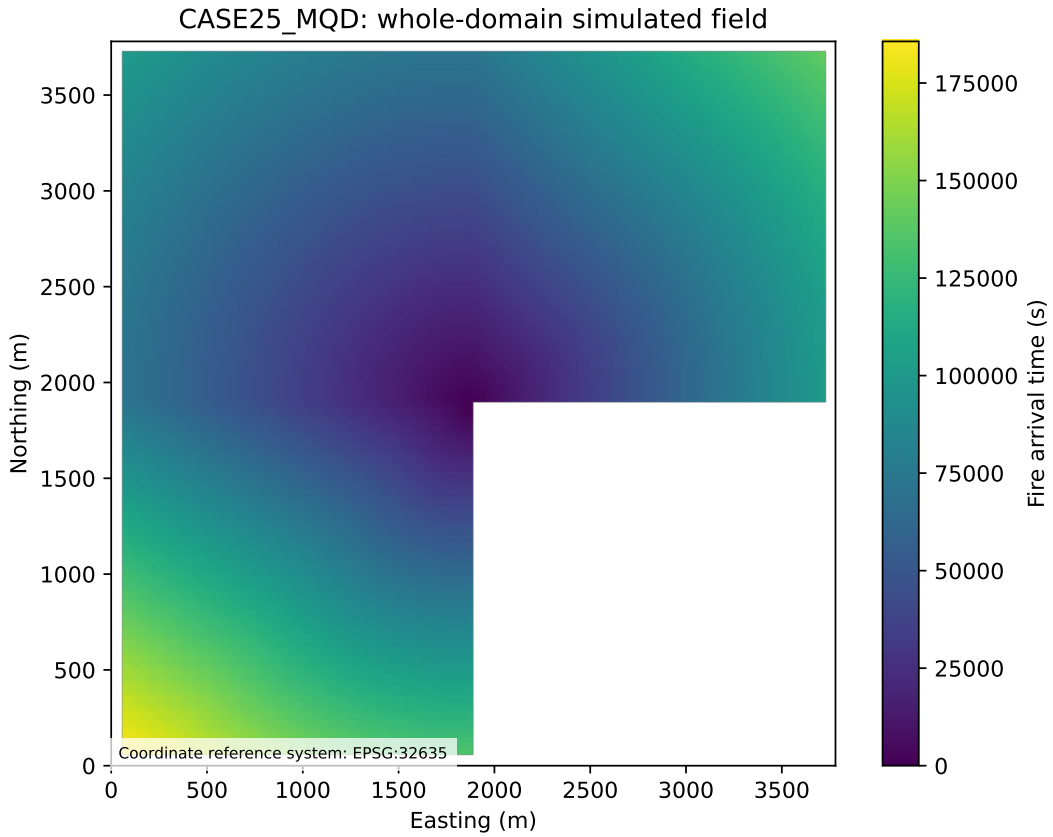


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / ROS @ 6%	1.511 m/min	1.51183 m/min	PASS
main / ROS @ 12%	1.089 m/min	1.08972 m/min	PASS
main / ROS @ 18%	0.801 m/min	0.801066 m/min	PASS
main / ROS @ 25%	0 m/min	0 m/min	PASS

The displayed overall decision is **PASS**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

Nonmonotone behavior suggests moisture raster or damping errors; significant spread in the extinction quadrant is a direct failure. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

Only one fuel model and the one-hour dead-fuel moisture field are varied. This case provides implementation verification evidence only.